

Simulation Modeling and Analysis

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Model Verification and Validation

- One of the most difficult problems facing a simulation analyst is that of trying to determine whether a simulation model is an accurate representation of the actual system being studied, i.e., whether the model is *valid*.
- *Verification* is concerned with determining whether the “assumptions document” has been correctly translated into a computer “program,” i.e., debugging the simulation computer program.
- Although verification is simple in concept, debugging a large-scale simulation program is a difficult task due to the potentially large number of logical paths.

Model Validation

- Whether the model is valid *for the particular objectives of a study*.
- The following are some general perspectives on validation:
 - A simulation model of a complex system can only be an *approximation* to the actual system, no matter how much effort is spent on model building. There is no such thing as absolute model validity, nor is it even desired.
 - “All models are wrong, but some are useful.”
 - A simulation model should always be developed for a particular set of purposes. A model that is valid for one purpose may not be for another.
 - Each time a simulation model is being considered for a new application, its validity should be reexamined

Accreditation

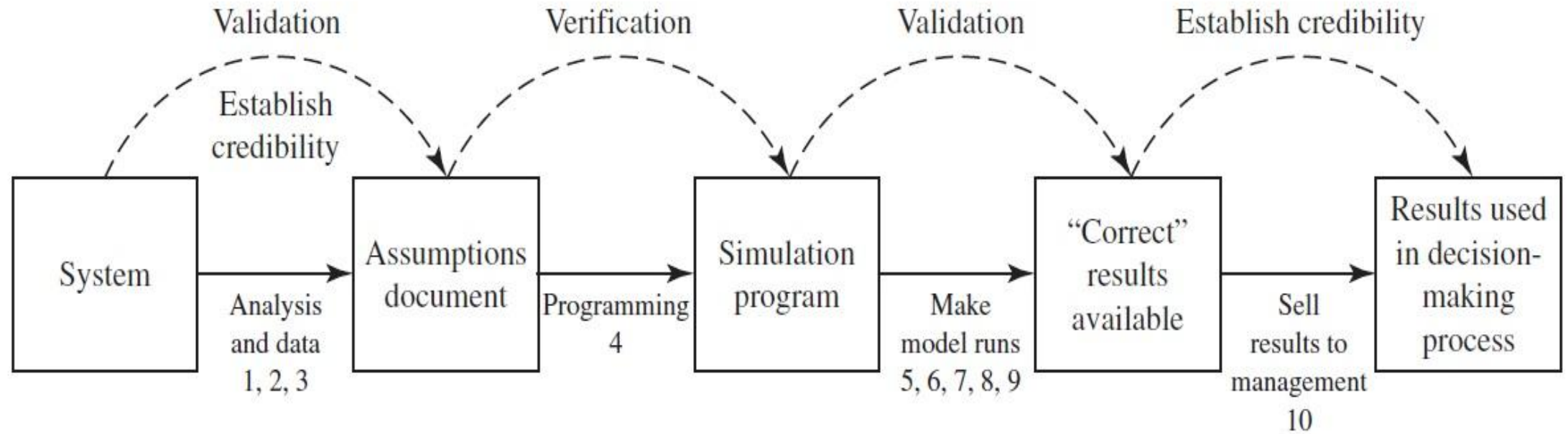
- The U.S. Department of Defense (DoD) is a large user of simulation models, and in recent years there has been considerable interest in verification, validation, and a concept known as accreditation (VV&A).
- Accreditation is the official certification (by the project sponsor) that a simulation model is acceptable for a specific purpose.
- The main reason that accreditation is mandated within DoD is that someone must take responsibility for the decision to use a model for a particular application, since a large amount of money and people's lives may be at stake.
- Also, most military analyses are done with legacy models, which may have been developed for another application or by another organization.

Accreditation

- Issues that are considered in an accreditation decision include:
 - Verification and validation that have been done
 - Simulation model development and use history (e.g., model developer and similar applications)
 - Quality of the data that are available
 - Quality of the documentation
 - Known problems or limitations with the simulation model
 - Credibility of the model

Verification and validation

- The timing and relationships of verification, validation and establishing credibility are shown in the next figure.
- The rectangles represent states of the model or the system of interest, the solid horizontal arrows correspond to the actions necessary to move from one state to another, and the curved dashed arrows show where the three major concepts are most prominently employed.
- The numbers below each solid arrow correspond to the steps in a sound simulation study, as discussed previously.



GUIDELINES FOR DETERMINING THE LEVEL OF MODEL DETAIL

- A simulation practitioner must determine what aspects of a complex real-world system actually need to be incorporated into the simulation model and at what level of detail, and what aspects can be safely ignored.
- It is rarely necessary to have a one-to-one correspondence between each element of the system and each element of the model.
- Modeling each aspect of the system will seldom be required to make effective decisions, and might result in excessive model execution time, in missed deadlines, or in obscuring important system factors.

- We now present some general guidelines for determining the level of detail required by a simulation model:
 - Carefully define the specific issues to be investigated by the study and the measures of performance that will be used for evaluation. Models are not universally valid, but are designed for specific purposes.
 - If the issues of interest have not been delineated, then it is impossible to determine the appropriate level of model detail. Since some models can accurately estimate one measure of performance but not another, it is also important to specify the performance measures of interest.
 - For example, a simple model of a manufacturing system might accurately predict throughput (e.g., parts per day) but be inadequate for determining the required floor space for work-in-process.
 - Finally, it is important to understand the manager's needs. A great model for the wrong problem will never be used.

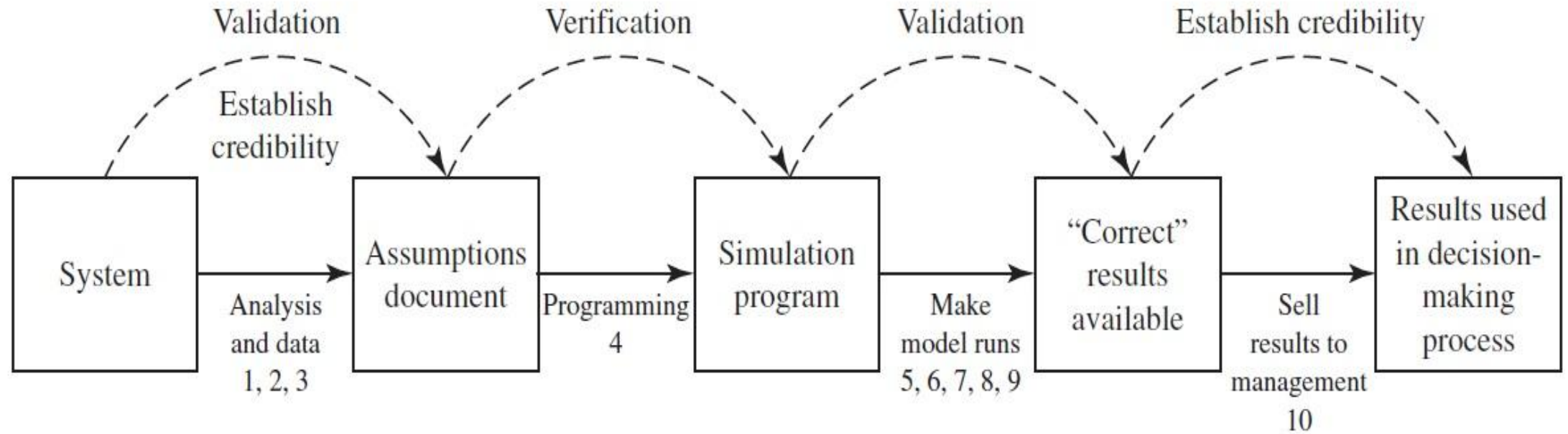
- A U.S. military analyst worked on a simulation model for six months without interacting with the general who requested it. At the Pentagon briefing for the study, the general walked out after 5 minutes stating, “That’s not the problem I’m interested in.”
- The entity moving through the simulation model does not always have to be the same as the entity moving through the corresponding system. Furthermore, it is not always necessary to model each component of the system in complete detail.
- Use subject-matter experts (SMEs) and sensitivity analyses to help determine the level of model detail. People who are familiar with systems similar to the one of interest are asked what components of the proposed system are likely to be the most important and, thus, need to be carefully modelled.

- A mistake often made by beginning modelers is to include an excessive amount of model detail. As a result, we recommend starting with a “moderately detailed” model, which can later be embellished if needed. The adequacy of a particular version of the model is determined in part by presenting the model to SMEs and managers. Regular interaction with these people also maintains their interest in the simulation study.
- If the number of factors (aspects of interest) for the study is large, then use a “coarse” simulation model to identify what factors have a significant impact on system performance. A “detailed” simulation model is then built, emphasizing those factors.

VERIFICATION OF SIMULATION COMPUTER PROGRAMS

- Technique 1: In developing a simulation model, write and debug the computer program in modules or subprograms.
- Technique 2: It is advisable in developing large simulation models to have more than one person review the computer program.
- Technique 3: Run the simulation under a variety of settings of the input parameters, and check to see that the output is reasonable.
- Technique 4: One of the most powerful techniques that can be used to debug a discrete-event simulation program is a “trace.” In a trace, the state of the simulated system are displayed just after each event occurs and are compared with hand calculations to see if the program is operating as intended

- Technique 5: The model should be run, when possible, under simplifying assumptions for which its true characteristics are known or can easily be computed.
- Technique 6: With some types of simulation models, it may be helpful to observe an animation of the simulation output.
- Technique 7: Compute the sample mean and sample variance for each simulation input probability distribution, and compare them with the desired (e.g., historical) mean and variance. This suggests that values are being correctly generated from these distributions.
- Technique 8: Use a commercial simulation package to reduce the amount of programming required.



Calibration and validation of models: Naylor and Finger Simulation Model Validation

- Naylor and Finger proposed three step approach for validation process. These steps are as follows:
 1. Face Validity
 2. Validate Model Assumptions
 3. Input - Output Transformation

Face Validity

- A model should appear reasonable on its face to model users and to those who know about the real system that is being simulated. (Cisco Packet Tracer)
- A model should be designed with high degree of realism regarding system structure and behaviour through reliable data.
- The potential users should also be involved in the validation process to aid in identification of model deficiencies and optimizing those deficiencies to produce a better model.
- Sensitivity analysis is also used for face validity of the model. It analyses the effect on output when there is change in input parameters.
- Sensitivity analysis is done through appropriate statistical techniques.

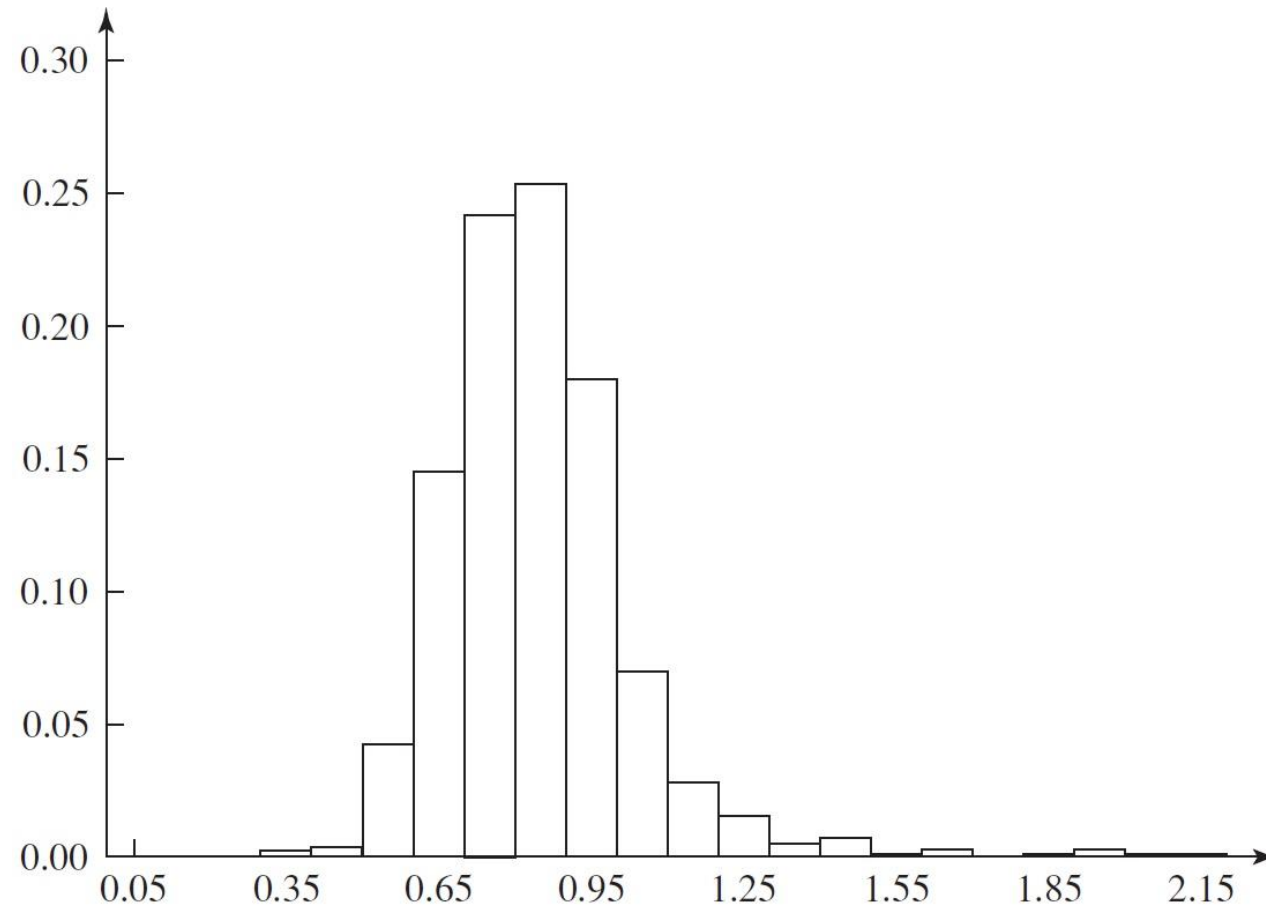
Validate Model Assumptions

- There are many assumptions that are made during model building process.
- These assumptions are of two types: structural assumption and data assumption.
- Structural assumptions involves simplification and abstraction of reality.
- Data assumptions should be done based on collection of reliable data.
- For example - Consider a bank system in which customer are queued before providing service.
- The structural assumptions may be - Customers form a single queue which is served by multiple tellers - Customers form one line for each teller - The number of teller should be fixed or variable.

Validate Model Assumptions

- These structural assumptions should be validated by actual observation during appropriate time periods and also by discussions with the managers and tellers.
- The data assumptions may be - interarrival times of customers during peak hour - interarrival times of customers during slack period - service time for personal accounts and so on.
- These data assumptions should be validated by consultation with bank managers.

A histogram for 856 ship-loading times (in days)



Input-Output Transformations

- It involves validating whether the model can predict the future behaviour of the real system when the model input data match the real inputs.
- In this phase, the model accepts values of the input parameters and transforms them into output measures of performance.
- The modeler uses the historical data reserved for validation purposes only.
- If in future, the model is used for different purpose, it should be revalidated in terms of new response of interest.
- For the complete input-output validation, at least one set of input conditions should be collected from live system so as to compare to model prediction.

Iterative Calibration for Validation

- Iterative calibration means to validate the model with the real system, look out for the places for betterment of the models and revising the model repeatedly until a satisfiable version is not achieved.
- The initial model is developed and is calibrated using Naylor-Finger calibration steps with the real system. It is then revised and a first revision model is generated.
- The first revision model is then calibrated with the real system. It is revised to form a second revision model.
- This process is continued until the model becomes acceptable.

